Fanqi Lin

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Education

Northwestern University **Sep 2025 – Expected Jun 2027**
Master of Science in Computer Engineering (MSCE) *Evanston, Illinois, U.S.*

Nanjing University **Sep 2021 – Jun 2025**
B.S. in Computer Science, Kuangyaming Honors School *Nanjing, Jiangsu, China*

University of California, Berkeley **Jan 2024 – May 2024**
Exchange Student in Computer Science *Berkeley, California, U.S.*

Technical Skills

Languages: Python, JavaScript/TypeScript (Learning), C/C++, Go, SQL
Web & Backend: **RESTful APIs**, **Microservices**, GraphQL (Familiar), React (Familiar)
Infrastructure: AWS, Docker, Kubernetes, PostgreSQL, Git

Experience

Nanjing Research Institute **Jun 2024 – Aug 2024**
AI Engineer Intern *Nanjing, China*

- **System Performance Verification:** Conducted cross-language functional audits between C-based high-performance kernels and Java-based management layers for a Network-level Load Balancing (NSLB) system.
- **Consistency Analysis:** Identified and resolved implementation discrepancies in GPU load distribution logic, ensuring system reliability during large-scale AI training tasks.
- **Code Quality Engineering:** Optimized the code review workflow by implementing segment-based analysis, reducing the manual review overhead for large-scale repository updates.

Project

GPU-Accelerated Stereo Vision & Depth Estimation System **Jan 2026 – Present**

- Designing a high-performance stereo matching pipeline to compute 3D depth maps from 2D image pairs using **CUDA** and C++.
- Parallelizing the **Stereo Rectification** and patch-matching processes by offloading computationally intensive cost calculations (SAD) to GPU kernels.
- Optimizing GPU throughput by implementing **Coalesced Memory Access** patterns and leveraging shared memory to minimize global memory latency.
- Conducting runtime performance profiling to benchmark **CUDA kernels** against serial CPU baselines across varying window sizes and resolutions.

Legal-RAG Agent: Standardized Benchmark & Efficient LLM System **Nov 2025 – Present**

- Engineered a standardized evaluation oracle bridging the **LegalAgentBench** dataset with the **A2A protocol**, ensuring interoperability and reproducible scoring for legal LLM agents.
- Architected a **Hybrid-RAG system** using **Voyage 3 Large Embeddings** and Matryoshka Representation Learning, achieving an **85% reduction in token usage** while supporting 32k context windows.
- Developed **HalluGraph**, a novel graph-based audit system that maps agent reasoning steps against a Ground Truth Knowledge Graph to detect subtle semantic hallucinations.
- Implemented **Self-Reflective Reasoning** combining **Chain-of-Thought (CoT)** and Self-RAG, enabling the agent to autonomously critique and validate legal citations before final output.

Voice-Activated Intelligent Recipe Assistant **Nov 2025 – Dec 2025**

- Designed a microservices-based architecture consisting of 3 decoupled components (Voice Client, Rasa Server, Action Server), communicating via RESTful APIs to ensure modularity and scalability.
- Engineered a robust backend handling 6 distinct query categories, including navigation, parameter extraction, and external API integration, fulfilling complex user requirements with 100% functional coverage.
- Resolved critical cross-platform C-library conflicts (PortAudio) on macOS, reducing environment setup time by 80% through a custom Conda configuration and environment.yml.

High-Performance Computing and Parallel Systems Development **Sep 2025 – Dec 2025**

- Designed and implemented a multi-process matrix multiplier using `ProcessPoolExecutor`, achieving near-linear speedup ($\approx$ **3.8**$\times$ on 4 cores) via efficient task partitioning.
- Developed a concurrent BFS algorithm, engineering **Mutexes** and **Atomic Operations** for thread safety, demonstrating parallel efficiency of over **75%**.
- Optimized image kernels using OpenMP `reduction` to solve a race condition, achieving an overall performance improvement of $\approx$ **5**$\times$ over the sequential baseline.

Secure File Sharing System **Feb 2024 - Mar 2024**

- Built a resilient file-sharing product with robust encryption and HMAC verification, ensuring data integrity in **production-grade scenarios**[cite: 34].
- Designed sophisticated permission management APIs to handle dynamic user revocation and access control.

Operating System & Kernel Development **Mar 2023 – Jun 2023**

- Developed a **Unix-like kernel** with page-based memory management, handling page faults and disk-RAM swapping.
- Implemented **kernel-level threads** and sync primitives (Mutexes) to support preemptive multitasking on RISC-V.